

1-8 b) $F(x,y,z) = \sum(1,2,3,6,7)$

x\y\z	00	01	11	10
0	0	1	1	0
1	0	1	1	1

$F(x,y,z) = xz + y$

c) $F(x,y,z) = \sum(4,5,6,7)$

x\y\z	00	01	11	10
0	0	0	0	0
1	1	1	1	1

$F(x,y,z) = xz + yz + xy$

1-9 b) $F(A,B,C,D) = \sum(3,7,11,12,14,15)$

AB\CD	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	0	1	1	1
10	0	1	1	1

$F(A,B,C,D) = AB + BCD + ACD$

d) $F(A,B,C,D) = \sum(0,2,4,5,6,7,8,10,13,15)$

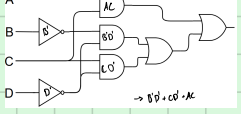
AB\CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$F(A,B,C,D) = B'C + BD + B'D$

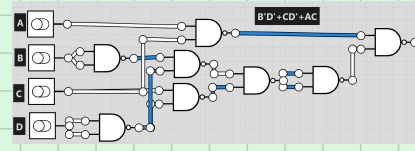
1-11) $F(A,B,C,D) = \sum(0,2,8,9,10,13,14,15)$

CD\AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$F(A,B,C,D) = B'D + C'D + AC$



b) same as 'a', but with only NAND gates

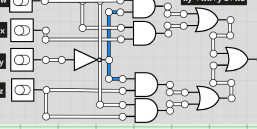


1-11. $F(w,x,y,z) = \sum(2,4,5,6,7,11,14,15)$

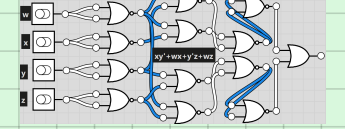
y\w\z	00	01	11	10
0	0	1	1	1
1	1	1	1	1
2	1	1	1	1

$F(w,x,y,z) = w'y + w'x + yz + wz$

c) 4-input logic diagram



d) 4-input logic



1-13. $F(w,x,y,z) = \sum(0,1,2,3,7,8,10)$

w\y\z	00	01	11	10
0	1	1	1	1
1	1	1	1	1
2	1	1	1	1

$F(w,x,y,z) = x'y + x'y'$

w\y\z	00	01	11	10
0	1	1	1	1
1	1	1	1	1
2	1	1	1	1

$F(w,x,y,z) = (A+B+C'D) \cdot (A+B'C'D) \cdot (A+B+C'D) \cdot (A+B'C'D) \cdot (A+B+C'D) \cdot (A+B'C'D)$

8. a) $F(x,y) = x'y + y'y + x'y'$

x\y	0	1
0	0	1
1	1	1

$F(x,y) = x'y + y$

b) $F(A,B,C) = A'B + AB'$

AB\C	0	1
0	1	0
1	0	1

$F(A,B,C) = A'B + C'$

10. Simplify

CD\AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$F(A,B,C,D) = A'C' + B'C' + A'D + BD$

12. a) $F(x,y) = x'y + y'y + x'y'$

i) $SOP = x'y + y$
 ii) $POS = (x+y')(x'+y)$

b) $F(A,B,C) = (A+B'D)(A+B'C)(A+B'D)(B+C'D)$

i) $SOP = A'B + C'D$
 ii) $POS = (A+C)(B'D)$

11. Simplify

a) $F(x,y,z) = \sum m(5,6,7)$

$F(x,y,z) = yz + xz'$

b) $F(x,y,z) = \sum m(0,4,6,7)$

$F(x,y,z) = z' + xy'$

c) $F(w,x,y,z) = \sum m(0,1,2,4,5,6,8,9,12,13,14)$

$F(w,x,y,z) = z' + w'z' + xz'$

d) $F(A,B,C,D) = \sum m(1,3,7,11,15)$

$F(A,B,C,D) = CD + AB'$

e) $F(A+B,y,z) = \sum m(1,2,5,6,12,14,15)$

$F(w,x,y,z) = x'z' + w'y'z + wxy$